

Credal Mixtures in Flow Based Variational Inference: Preliminary Theoretical and Empirical Results

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Problem Setting and Hypothesis

In Bayesian inference under prior uncertainty, one is often led to consider a finite family of plausible posteriors $\{p_k\}_{k=1}^K$, induced by different priors or hyperparameters. Suppose P denotes the data generating distribution. The functional

$$q \mapsto \text{KL}(P\|q)$$

is convex in q . Consequently, for any candidate approximations $\{q_k\}_{k=1}^K$ and any convex mixture

$$q_w(x) = \sum_{k=1}^K w_k q_k(x), \quad w \in \Delta^{K-1},$$

the optimal mixture satisfies

$$\min_{w \in \Delta^{K-1}} \text{KL}(P\|q_w) \leq \min_{k \in \{1, \dots, K\}} \text{KL}(P\|q_k).$$

The central hypothesis motivating this work is that this convexity based guarantee should be empirically visible for *flow based variational posteriors*, provided that mixture selection is performed with a likelihood consistent objective and that estimation noise is controlled.

Initial Empirical Obstruction

In initial experiments, each posterior approximation q_k was learned independently using flow matching under different priors. Credal mixtures were then formed either by mixing samples or by selecting weights using sample based proxies for divergence (for example MMD or adversarial KL bounds).

In this setting, credal mixtures frequently underperformed the best single posterior under held out negative log likelihood. This behaviour does not contradict convexity of $\text{KL}(P\|q)$, because the convexity guarantee concerns the exact mixture density q_w and the exact KL objective. In practice, we were solving a different problem due to estimation and optimisation effects:

- Mixture weights were selected using noisy or indirect objectives rather than the KL to P (approximated by held out NLL).
- Sample based estimates introduced variance that could mis rank mixtures, particularly when differences were small.
- When a mixture was approximated implicitly (for example via resampling or finite sample evaluation), the resulting objective was not a faithful estimate of $\text{KL}(P\|q_w)$.

These effects can mask the convexity predicted behaviour even when the underlying guarantee holds in principle.

Architectural Resolution

To reduce estimation noise and ensure likelihood consistent mixture selection, the architecture and training procedure were modified to enforce a coherent variational family across priors:

- A single flow matching model $v_\theta(x, t, p)$ was trained, conditioned on a discrete prior index p .
- All priors shared the same backbone network and time embedding, ensuring comparable approximation capacity and training dynamics across k .
- Mixture weights were optimised by directly minimising held out negative log likelihood, a Monte Carlo approximation to $\text{KL}(P||q)$ up to an additive constant.

This construction makes mixture selection better conditioned, because the objective aligns with the convexity guarantee and the family $\{q_k\}$ is trained within a shared optimisation and modelling framework.

Empirical Results

Experiments were performed on low dimensional synthetic benchmarks, specifically an eight component Gaussian ring distribution, a two moons distribution, and a two arm spiral distribution, using 10 random seeds per dataset. Performance was evaluated using held out negative log likelihood estimated via Monte Carlo sampling.

Across all datasets, the optimised credal mixture q_{w^*} achieved performance close to the best single q_k . In these runs, the mixture showed a small mean degradation in held out NLL relative to the best single component (between 0.015 and 0.053), with seed to seed variability of comparable magnitude. These preliminary results suggest that likelihood consistent mixture optimisation and variance control remove the larger failure modes seen in earlier pipelines, but do not yet yield consistent improvements over the best single posterior.

Summary of empirical results.

Dataset	Seeds	Best single NLL (mean $\pm$ std)	Mixture NLL (mean $\pm$ std)
Eight Gaussian ring	10	1.940 ± 0.047	1.993 ± 0.076
Two moons	10	1.356 ± 0.015	1.371 ± 0.027
Two arm spirals	10	3.081 ± 0.017	3.114 ± 0.040

While the mixture does not improve upon the best single posterior in these settings, it remains close in absolute performance. A more decisive comparison requires paired, per seed analysis and tighter control of Monte Carlo evaluation error, which is part of ongoing work.

Experimental Setup (Summary)

Posterior approximations were learned using continuous time flow matching with a shared velocity field backbone. Conditioning on the prior index p was implemented via a learned embedding concatenated with the time embedding and injected into all network blocks.

Negative log likelihood was estimated by integrating the learned flow using a fixed step ODE solver and evaluating the resulting density via Monte Carlo samples from a held out validation set. Credal mixture weights were optimised on the probability simplex using a softmax parameterisation and gradient based optimisation of held out negative log likelihood.

Interpretation and Research Contribution

The primary insight from these experiments is that the convexity guarantee is not automatically reflected by modern training pipelines unless the mixture selection objective matches the theory and the estimation procedure has low enough variance.

In particular, the results indicate that:

- mixture selection should be likelihood consistent (held out NLL rather than proxy divergences),
- estimator variance must be tracked and reported (mean and standard deviation across seeds, and ideally paired comparisons),
- and deviations from the convexity based guarantee in practice can often be attributed to optimisation and estimation, not the underlying theorem.

This reframes the problem from asking whether credal mixtures work with flows to understanding when classical convexity arguments remain empirically visible under deep variational approximations.

Ongoing and Future Work

Current work focuses on:

- formalising sufficient conditions under which mixture selection is stable for deep variational families,
- extending the framework from discrete priors to continuous credal sets,
- improving evaluation by reporting paired, per seed deltas and quantifying Monte Carlo error in likelihood estimation.

All results reported here should be viewed as preliminary, with full reproducible experiments and broader empirical evaluation forming the next stage of this research.